

Mohamed Abdelrahman

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EDUCATION

University of Houston

Bachelor of Science in Computer Science

- GPA: 3.28 / 4.00

Houston, TX

Aug. 2023 – May 2026

EXPERIENCE

Code[Coogs] Competitions Director

Jan 2025 – May 2025

University Of Houston

Houston, TX

- Led 9 competitive programming events including weekly workshops and contests using C++ to teach data structures, graph algorithms, and complexity analysis to 30+ students.
- Organized and hosted algorithmic contests on Kattis, curating problem sets aligned with LeetCode-style technical interview topics and CS fundamentals.
- Engineered interactive workshop formats with live coding, peer discussions, and real-time feedback loops to improve engagement and learning outcomes.
- Collaborated with student leaders and faculty to expand outreach, increase attendance, and iterate on event formats based on user-centered feedback.

Coding Expert - AI Trainer (Contract)

Aug 2023 – Jan 2025

Outlier.ai

Remote

- Developed solutions for fine-tuning code generation models using Python and C++, improving quality by 40%.
- Created prompts to produce model deviations that improve model training using prompt engineering techniques.
- Annotated LLM output for over 400 responses and rated its quality based on a rigorous rubric.
- Reviewed submissions for other contributors and provided feedback, improving their task quality by 30%.

Software Engineering Intern

Aug. 2022 – Jan 2023

bld.ai

Remote

- Collaborated with four interns to develop a full-stack application featuring a user interface, server, and database using React with JavaScript, Django with Python, and PostgreSQL
- Refactored database models, authorization, and data validation, reducing bugs in the codebase by 50%
- Deployed a minimum viable product (MVP) within one month, following a two-week sprint cycle
- Followed Agile Practices, including sprint planning, daily stand-ups, retrospectives, and iterative development to optimize team workflow and product delivery by 50%

PROJECTS

OrderedMap C++ Library | C++, STL, Smart Pointers, Custom Iterators, TDD

May 2025 – Present

- Designed and implemented a custom `OrderedMap<Key, Value>` data structure maintaining insertion order with fast key lookups
- Built on a doubly linked list with unique ownership via `std::unique_ptr` and custom bidirectional iterators
- Supported operations like `insert`, `erase`, `move_to_front`, and `move_to_back` with iterator stability
- Developed unit tests using Catch2 and followed Test-Driven Development (TDD) practices
- Documented the project with diagrams and maintained a clean, modular design for open-source use

Nota | Flutter, Dart, Bloc, SQFlite, Quill, Material 3

Jan 2025 – Present

- Developed a rich-text note-taking app for Android with offline storage using SQL queries in SQFlite.
- Integrated Flutter Quill editor to support formatting such as bold, italics, headers, links, and lists.
- Implemented labeling and filtering features for efficient note organization and retrieval.
- Managed UI and business logic using Bloc state management for clean separation of concerns.
- Designed a responsive UI with Material 3 components and support for light/dark theme switching.

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript, HTML/CSS, Dart

Frameworks: Django, React, NextJS, Flutter

Developer Tools: Git, Docker, Postman, VS Code, CLion, Visual Studio, PyCharm, IntelliJ, DBeaver

Libraries: unittest, React Router, Bloc, Flutter Quill, SQFlite, Axios